

FM Exam Notes

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Chapter 1

Interest Theory

Chapter 2

Annuities

2.1 Basic Annuities

Sum of a Geometric Series

Before deriving annuity formulas, we recall the sum of a finite geometric series.

Consider the sum

$$S = a + ar + ar^2 + \cdots + ar^{n-1}.$$

Multiply both sides by r :

$$rS = ar + ar^2 + ar^3 + \cdots + ar^n.$$

Now subtract the second equation from the first:

$$S - rS = (a + ar + ar^2 + \cdots + ar^{n-1}) - (ar + ar^2 + \cdots + ar^{n-1} + ar^n).$$

All middle terms cancel, leaving

$$S - rS = a - ar^n.$$

Thus,

$$S(1 - r) = a - ar^n.$$

Therefore, if $r \neq 1$,

$$S = \frac{a - ar^n}{1 - r}.$$

Equivalently,

$$a + ar + ar^2 + \cdots + ar^{n-1} = \frac{a(1 - r^n)}{1 - r}.$$

Finite Geometric Series

For $r \neq 1$,

$$a + ar + ar^2 + \cdots + ar^{n-1} = \frac{a(1 - r^n)}{1 - r}.$$

In words, the sum of a geometric series is

$$\frac{\text{first term} - \text{next term after the last term}}{1 - \text{common ratio}}.$$

Special Case

The formula above assumes that $r \neq 1$.

If $r = 1$, then every term is equal to a , so

$$S = a + a + \cdots + a = na.$$

Using the Geometric Series Formula

Given $i = 0.05$, find

$$v^2 + v^4 + \cdots + v^{12}.$$

Solution

Since

$$v = \frac{1}{1+i},$$

we have

$$v = \frac{1}{1.05}.$$

The expression

$$v^2 + v^4 + \cdots + v^{12}$$

is a finite geometric series.

The first term is

$$a = v^2,$$

the common ratio is

$$r = v^2,$$

and the next term after the last term is

$$v^{14}.$$

Using the geometric series formula,

$$S = \frac{\text{first term} - \text{next term after the last term}}{1 - \text{common ratio}},$$

we get

$$v^2 + v^4 + \cdots + v^{12} = \frac{v^2 - v^{14}}{1 - v^2}.$$

Substitute $v = 1.05^{-1}$:

$$\frac{v^2 - v^{14}}{1 - v^2} = \frac{1.05^{-2} - 1.05^{-14}}{1 - 1.05^{-2}}.$$

Therefore,

$$v^2 + v^4 + \cdots + v^{12} = 4.32.$$

Hence,

$$\boxed{4.32}.$$

A More Complicated Geometric Series

Given $i = 0.08$ and $r = 0.05$, find

$$(1+i)^{20} + (1+i)^{25}(1+r)^3 + (1+i)^{30}(1+r)^6 + \cdots + (1+i)^{55}(1+r)^{21}.$$

Solution

This is still a geometric series. The first term is

$$a = (1+i)^{20}.$$

To move from one term to the next, the exponent of $(1+i)$ increases by 5, and the exponent of $(1+r)$ increases by 3.

Therefore, the common ratio is

$$q = (1+i)^5(1+r)^3.$$

The last term is

$$(1+i)^{55}(1+r)^{21}.$$

The next term after the last term is

$$(1+i)^{60}(1+r)^{24}.$$

Using the geometric series formula,

$$S = \frac{\text{first term} - \text{next term after the last term}}{1 - \text{common ratio}},$$

we get

$$S = \frac{(1+i)^{20} - (1+i)^{60}(1+r)^{24}}{1 - (1+i)^5(1+r)^3}.$$

Substitute $i = 0.08$ and $r = 0.05$:

$$S = \frac{1.08^{20} - (1.08)^{60}(1.05)^{24}}{1 - (1.08)^5(1.05)^3}.$$

Therefore,

$$S = 459.25.$$

Hence,

$$\boxed{459.25}.$$

2.1.1 Annuity-Immediate

What is an Annuity-Immediate?

Annuity

An **annuity** is a series of payments made at equal time intervals.

Common examples include rent payments, mortgage payments, car payments, paychecks, and pension benefits.

Annuity-Immediate

An **annuity-immediate** is an annuity whose payments are made at the **end** of each payment period.

For example, a 10-year annuity-immediate with annual payments has payments at times

$$1, 2, 3, \dots, 10.$$

There is no payment at time 0.

Present Value of an Annuity-Immediate

Given $i = 0.05$, find the present value of a 10-year annuity-immediate with annual payments of 100.

Solution

Since this is an annuity-immediate, the payments occur at the end of each year.

Thus, the payments of 100 occur at times

$$1, 2, 3, \dots, 10.$$

Let

$$v = \frac{1}{1+i}.$$

Since $i = 0.05$,

$$v = \frac{1}{1.05}.$$



The present value is the sum of the present values of all payments:

$$PV = 100v + 100v^2 + \dots + 100v^{10}.$$

This is a geometric series. The first term is

$$100v,$$

the common ratio is

$$v,$$

and the next term after the last term is

$$100v^{11}.$$

Using the geometric series formula,

$$PV = \frac{100v - 100v^{11}}{1 - v}.$$

Substitute

$$v = \frac{1}{1.05}.$$

Then

$$PV = \frac{100(1.05)^{-1} - 100(1.05)^{-11}}{1 - (1.05)^{-1}}.$$

Therefore,

$$PV = 772.17.$$

Hence,

$$\boxed{PV = 772.17}.$$

Present Value

We now derive the present value formula for an annuity-immediate from the geometric series formula introduced above.

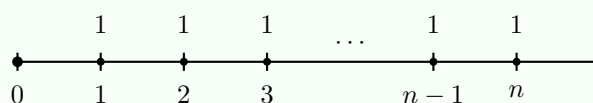
Present Value of an Annuity-Immediate

At an annual effective interest rate i , the present value of an n -year annuity-immediate with annual payments of 1 is denoted by

$$a_{\overline{n}|}.$$

The payments are made at the end of each year, so they occur at times

$$1, 2, 3, \dots, n.$$



Since the payment at time k has present value v^k , the present value is

$$PV = v + v^2 + \dots + v^n.$$

This is a finite geometric series with first term v , common ratio v , and next term after the last term v^{n+1} . Therefore,

$$PV = \frac{v - v^{n+1}}{1 - v}.$$

Factor v in the numerator:

$$PV = \frac{v(1 - v^n)}{1 - v}.$$

Now recall that

$$v = \frac{1}{1 + i}.$$

Thus,

$$1 - v = 1 - \frac{1}{1 + i} = \frac{i}{1 + i} = iv.$$

Therefore,

$$PV = \frac{v(1 - v^n)}{iv}.$$

Cancel v :

$$PV = \frac{1 - v^n}{i}.$$

Hence,

$$\boxed{a_{\overline{n}|} = \frac{1 - v^n}{i}}.$$

Meaning of $a_{\overline{n}|}$

The symbol

$$a_{\overline{n}|}$$

represents the present value of n payments of 1, paid at the end of each period.
If each payment is R instead of 1, then the present value is

$$Ra_{\overline{n}|}.$$

So the coefficient of $a_{\overline{n}|}$ is the amount of each payment.

Present Value of a Basic Annuity-Immediate

Given $i = 0.05$, find the present value of a 10-year annuity-immediate with annual payments of 100.

Solution

A 10-year annuity-immediate has payments at times

$$1, 2, 3, \dots, 10.$$

Since each payment is 100, the present value is

$$PV = 100a_{\overline{10}|}.$$

Using

$$a_{\overline{n}|} = \frac{1 - v^n}{i},$$

we get

$$PV = 100 \left(\frac{1 - v^{10}}{0.05} \right).$$

Since

$$v = \frac{1}{1.05},$$

we have

$$PV = 100 \left(\frac{1 - (1.05)^{-10}}{0.05} \right).$$

Therefore,

$$PV = 772.17.$$

Hence,

$$\boxed{PV = 772.17}.$$

Where $a_{\overline{n}|}$ Is Measured

The symbol

$$a_{\overline{n}|}$$

represents the value of n payments of 1, measured **one period before the first payment**.
So if payments occur at times

$$1, 2, \dots, n,$$

then $a_{\overline{n}|}$ is measured at time 0.

If payments occur at times

$$4, 5, 6, 7,$$

then $a_{\overline{4}|}$ is first measured at time 3, one period before the first payment. To find the value at time 0, we must discount by v^3 .

A Shifted Annuity-Immediate

Given $i = 0.10$, find the present value of payments of 200 at times 4, 5, 6, and 7.

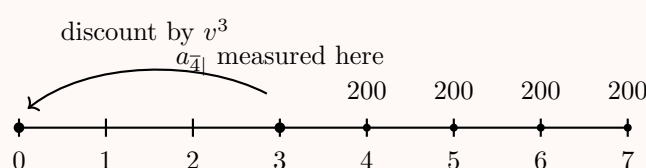
Solution

The payments are

$$200, 200, 200, 200$$

at times

$$4, 5, 6, 7.$$



The four payments form an annuity-immediate of 4 payments, but the first payment occurs at time 4.

Therefore,

$$200a_{\overline{4}|}$$

is measured at time 3, one period before the first payment.

To find the present value at time 0, we discount by v^3 :

$$PV = 200a_{\overline{4}|}v^3.$$

Using

$$a_{\overline{4}|} = \frac{1 - v^4}{i},$$

we get

$$PV = 200 \left(\frac{1 - v^4}{0.10} \right) v^3.$$

Since

$$v = \frac{1}{1.10},$$

we have

$$PV = 200 \left(\frac{1 - (1.10)^{-4}}{0.10} \right) (1.10)^{-3}.$$

Therefore,

$$PV = 476.31.$$

Hence,

$$\boxed{PV = 476.31}.$$

Alternatively, we can view the payments at times 4, 5, 6, 7 as the difference between a 7-year annuity-immediate and a 3-year annuity-immediate:

$$PV = 200a_{\overline{7}|} - 200a_{\overline{3}|}.$$

This gives the same result:

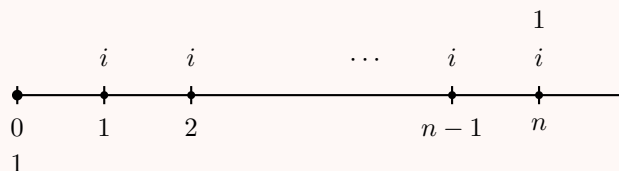
$$\boxed{PV = 476.31}.$$

Alternative Derivation of $a_{\overline{n}|}$

We can derive the present value formula for an annuity-immediate using a simple investment argument.

Suppose you invest 1 in a fund that pays an annual effective interest rate i .

At the end of each year, you withdraw only the interest earned during that year, leaving the principal 1 in the fund. After n years, you withdraw the principal 1.



The original investment has present value 1.

The interest payments form an n -year annuity-immediate with payments of i , so their present value is

$$ia_{\overline{n}|}.$$

The final withdrawal of the principal 1 at time n has present value

$$v^n.$$

Using an equation of value at time 0, we get

$$ia_{\overline{n}|} + v^n = 1.$$

Therefore,

$$ia_{\overline{n}|} = 1 - v^n.$$

Dividing both sides by i , we obtain

$$a_{\overline{n}|} = \frac{1 - v^n}{i}.$$

Thus,

$$\boxed{a_{\overline{n}|} = \frac{1 - v^n}{i}}.$$

Rate per Payment Period

In the formula

$$a_{\overline{n}|} = \frac{1 - v^n}{i},$$

the rate i must be interpreted as the **effective interest rate per payment period**.

Similarly, v is the discount factor per payment period:

$$v = \frac{1}{1 + i}.$$

For example, if payments are monthly, then i must be the effective monthly interest rate, not the annual effective rate.

Monthly Payments

Given an annual effective interest rate of

$$i = 0.061677812,$$

find the present value of a 5-year annuity-immediate with monthly payments of 100.

Solution

Since the payments are monthly, we need the effective monthly interest rate. Let j be the effective monthly rate.

Then

$$1 + j = (1.061677812)^{1/12}.$$

Thus,

$$j = (1.061677812)^{1/12} - 1.$$

Therefore,

$$j = 0.005.$$

The number of monthly payments is

$$5(12) = 60.$$

So the present value is

$$PV = 100a_{\overline{60}|j}.$$

Using the annuity-immediate formula,

$$a_{\overline{60}|j} = \frac{1 - v_j^{60}}{j},$$

where

$$v_j = \frac{1}{1 + j}.$$

Thus,

$$PV = 100 \left(\frac{1 - v_j^{60}}{j} \right).$$

Substitute $j = 0.005$:

$$PV = 100 \left(\frac{1 - (1.005)^{-60}}{0.005} \right).$$

Therefore,

$$PV = 5,172.56.$$

Hence,

$$\boxed{PV = 5,172.56}.$$

Exercise 2.1.1

A payment of 1 is to be made each year for 22 years.

The effective annual rate of interest is 8.25%. If the first payment is one year from now, the present value of these payments is 10.

Calculate the present value if the first payment is 22 years from now.

A. 1.69

- B. 1.79
- C. 1.89
- D. 1.99
- E. 2.09

Solution

If the first payment is one year from now, the payments occur at times

$$1, 2, \dots, 22.$$

We are given that the present value of this annuity-immediate is

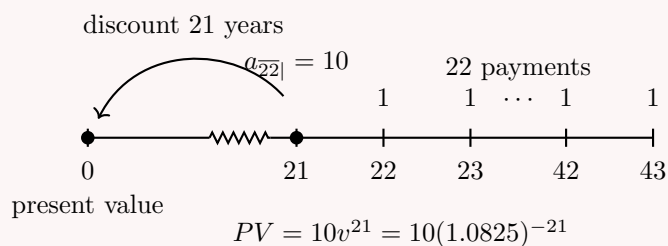
$$a_{\overline{22}|} = 10.$$

If the first payment is instead 22 years from now, then the payments occur at times

$$22, 23, \dots, 43.$$

This is still a 22-payment annuity-immediate, but its value is measured one period before the first payment. Therefore, the value of the annuity at time 21 is

$$a_{\overline{22}|} = 10.$$



Therefore, the present value at time 0 is

$$PV = 10v^{21}.$$

Since

$$i = 0.0825,$$

we have

$$v = \frac{1}{1.0825}.$$

Thus,

$$PV = 10(1.0825)^{-21}.$$

Therefore,

$$PV = 1.89.$$

Hence,

$$\boxed{PV = 1.89}.$$

The correct choice is

$$\boxed{\text{C}}.$$

Exercise 2.1.2

John invests 100,000 into a fund earning an annual effective interest rate of 4%. John will make withdrawals of X at the end of every 5 years for a total of 10 withdrawals. After the last withdrawal, the fund's balance will be 0. Calculate X .

- A. 23,213.09
- B. 24,213.09
- C. 25,213.09
- D. 26,213.09
- E. 27,213.09

Solution

The withdrawals are made every 5 years. Therefore, we should use the effective interest rate per 5-year period.

Let j be the effective rate per 5 years. Since the annual effective interest rate is 4%,

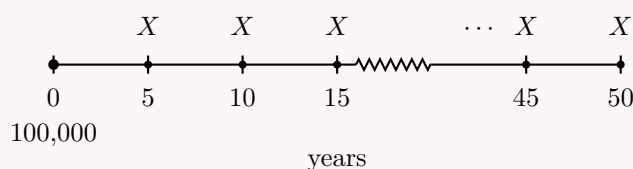
$$1 + j = (1.04)^5.$$

Thus,

$$j = (1.04)^5 - 1.$$

So

$$j = 0.216652902.$$



The withdrawals form an annuity-immediate with 10 payments of X , where each payment period is 5 years.

Thus, the present value of the withdrawals is

$$Xa_{\overline{10}|j}.$$

Since the initial fund is 100,000, the equation of value at time 0 is

$$100,000 = Xa_{\overline{10}|j}.$$

Using

$$a_{\overline{10}|j} = \frac{1 - v_j^{10}}{j},$$

we get

$$100,000 = X \left(\frac{1 - v_j^{10}}{j} \right),$$

where

$$v_j = \frac{1}{1 + j}.$$

Since

$$j = 0.216652902,$$

we have

$$v_j = \frac{1}{1.216652902}.$$

Therefore,

$$100,000 = X \left(\frac{1 - (1.216652902)^{-10}}{0.216652902} \right).$$

Solving for X ,

$$X = \frac{100,000}{\frac{1 - (1.216652902)^{-10}}{0.216652902}}.$$

Thus,

$$X = 25,213.09.$$

Hence,

$$\boxed{X = 25,213.09}.$$

The correct choice is

$$\boxed{\text{C}}.$$

Exercise 2.1.3

A 20-year annuity pays 100 every other year beginning at the end of the second year, with additional payments of 300 each at the ends of years 3, 9, and 15.

The effective annual interest rate is 4%.

Calculate the present value of the annuity.

- A. 1,310
- B. 1,340
- C. 1,370
- D. 1,400
- E. 1,430

Solution

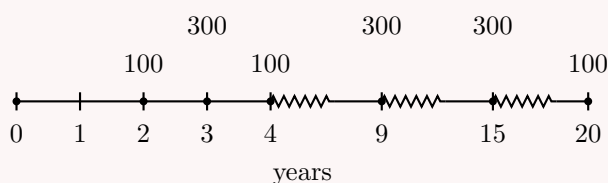
Let

$$v_{0.04} = \frac{1}{1.04}.$$

The annuity pays 100 every other year beginning at time 2. Therefore, the regular payments occur at times

$$2, 4, 6, \dots, 20.$$

There are 10 regular payments.



For the payments of 100, the payment period is 2 years. Therefore, we use the effective interest rate per 2-year period.

Let j be the effective rate per 2 years. Then

$$1 + j = (1.04)^2.$$

Thus,

$$j = (1.04)^2 - 1 = 0.0816.$$

The present value of the regular payments of 100 is

$$100a_{\overline{10}|j}.$$

The additional payments of 300 occur at times 3, 9, and 15. Their present value is

$$300v_{0.04}^3 + 300v_{0.04}^9 + 300v_{0.04}^{15}.$$

Therefore, the total present value is

$$PV = 100a_{\overline{10}|j} + 300v_{0.04}^3 + 300v_{0.04}^9 + 300v_{0.04}^{15}.$$

Using

$$a_{\overline{10}|j} = \frac{1 - v_j^{10}}{j},$$

we get

$$PV = 100 \left(\frac{1 - (1.0816)^{-10}}{0.0816} \right) + 300(1.04)^{-3} + 300(1.04)^{-9} + 300(1.04)^{-15}.$$

Thus,

$$PV = 1,310.25.$$

So the present value is approximately

$$\boxed{1,310}.$$

The correct choice is

$$\boxed{\text{A}}.$$

Exercise 2.1.4

On January 1, 1985, Marc has the following two options for repaying a loan:

- (i) sixty monthly payments of 100, commencing February 1, 1985;
- (ii) a single payment of 6,000 at the end of K months.

Interest is at a nominal annual rate of 12%, compounded monthly. The two options have the same present value.

Determine K .

- A. 29.0
- B. 29.5
- C. 30.0

D. 30.5

E. 31.0

Solution

The nominal annual interest rate is 12%, compounded monthly. Therefore, the effective monthly interest rate is

$$j = \frac{0.12}{12} = 0.01.$$

Let

$$v_{0.01} = \frac{1}{1.01}.$$

The first option consists of 60 monthly payments of 100, beginning one month from January 1, 1985. Therefore, its present value is

$$100a_{\overline{60}|0.01}.$$

The second option is a single payment of 6,000 at the end of K months. Its present value is

$$6,000v_{0.01}^K.$$

Since the two options have the same present value,

$$100a_{\overline{60}|0.01} = 6,000v_{0.01}^K.$$

Using

$$a_{\overline{60}|0.01} = \frac{1 - v_{0.01}^{60}}{0.01},$$

we get

$$100 \left(\frac{1 - v_{0.01}^{60}}{0.01} \right) = 6,000v_{0.01}^K.$$

Thus,

$$v_{0.01}^K = \frac{100 \left(\frac{1 - v_{0.01}^{60}}{0.01} \right)}{6,000}.$$

Substituting

$$v_{0.01} = \frac{1}{1.01},$$

we obtain

$$v_{0.01}^K = 0.7492506.$$

Taking natural logarithms,

$$K \ln(v_{0.01}) = \ln(0.7492506).$$

Therefore,

$$K = \frac{\ln(0.7492506)}{\ln(v_{0.01})}.$$

Since

$$v_{0.01} = \frac{1}{1.01},$$

we get

$$K = 29.0123.$$

Thus,

$$K \approx 29.0.$$

The correct choice is

A.

Exercise 2.1.5

An annuity pays 2 at the end of each year for 18 years.

Another annuity pays 2.5 at the end of each year for 9 years.

At an effective annual interest rate of i , where $0 < i < 1$, the present values of both annuities are equal.

Calculate i .

A. 14%

B. 17%

C. 20%

D. 23%

E. 26%

Solution

Let

$$v = \frac{1}{1+i}.$$

The present value of the first annuity is

$$2a_{\overline{18}|}.$$

The present value of the second annuity is

$$2.5a_{\overline{9}|}.$$

Since the present values are equal,

$$2a_{\overline{18}|} = 2.5a_{\overline{9}|}.$$

Using

$$a_{\overline{n}|} = \frac{1-v^n}{i},$$

we get

$$2 \left(\frac{1-v^{18}}{i} \right) = 2.5 \left(\frac{1-v^9}{i} \right).$$

Since $i > 0$, multiply both sides by i :

$$2(1-v^{18}) = 2.5(1-v^9).$$

Expand:

$$2 - 2v^{18} = 2.5 - 2.5v^9.$$

Rearrange:

$$2v^{18} - 2.5v^9 + 0.5 = 0.$$

Let

$$x = v^9.$$

Then

$$v^{18} = x^2.$$

So the equation becomes

$$2x^2 - 2.5x + 0.5 = 0.$$

Using the quadratic formula,

$$x = \frac{2.5 \pm \sqrt{2.5^2 - 4(2)(0.5)}}{2(2)}.$$

Thus,

$$x = 1 \quad \text{or} \quad x = 0.25.$$

Since

$$x = v^9,$$

we have

$$v^9 = 1 \quad \text{or} \quad v^9 = 0.25.$$

The solution $v^9 = 1$ would imply $v = 1$, hence $i = 0$. But the problem states that $0 < i < 1$, so we discard this solution.

Therefore,

$$v^9 = 0.25.$$

Taking the ninth root:

$$v = (0.25)^{1/9}.$$

Since

$$v = \frac{1}{1+i},$$

we have

$$1+i = \frac{1}{v}.$$

Thus,

$$i = \frac{1}{(0.25)^{1/9}} - 1.$$

Therefore,

$$i = 0.16653.$$

So

$$i \approx 16.653\%.$$

The closest answer choice is

$$\boxed{17\%}.$$

The correct choice is

$$\boxed{\text{B}}.$$

Useful Tools

Accumulated Value

2.1.2 Annuity-Due

Present Value

Accumulated Value

2.2 Varying Annuities

2.2.1 Geometric Varying Annuities

Part 1

Part 2

Chapter 3

Loans

3.1 Coming Soon

This chapter will contain loan repayment, outstanding balance, amortization method, sinking fund method, and amortization schedules.

Chapter 4

Bonds

4.1 Coming Soon

This chapter will contain bond pricing, premium and discount bonds, book value, amortization of premium, accumulation of discount, and callable bonds.

Chapter 5

Immunization and Duration

5.1 Coming Soon

This chapter will contain Macaulay duration, modified duration, convexity, full immunization, and Redington immunization.